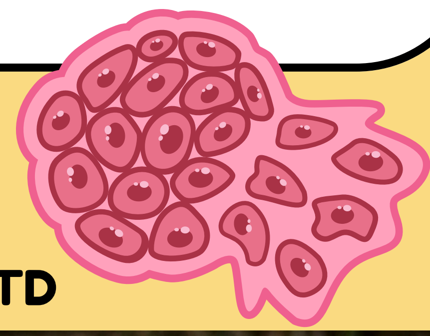
Devils within Devils: Decoding transmissible cancers

Leung Hoi Shun Bethanie, Yeang Anson,
Sim Ee Hean Matthieu, I Gusti Agung Ayu Risma Riyanti



What Are Clonally Transmissible Cancers?

- For a cancer to be **horizontally transmissible** as an **allograph**, the cell itself must be able to **migrate** between hosts
- Factors that allow for **inter-individual** transmission are what **differentiates normal** cancer cells and **clonally transmissible** cancer cells

4 main conditions necessary for **direct passage** of cancer cells between hosts:

1	2	3	4
Tumor tissue properties that **promote shedding** of large numbers of malignant cells	Some **interactions** between hosts that allow transmission	**Tumor cell plasticity** that permits survival during transmission and establishment in a new host	A "**permissible**" host or host tissue

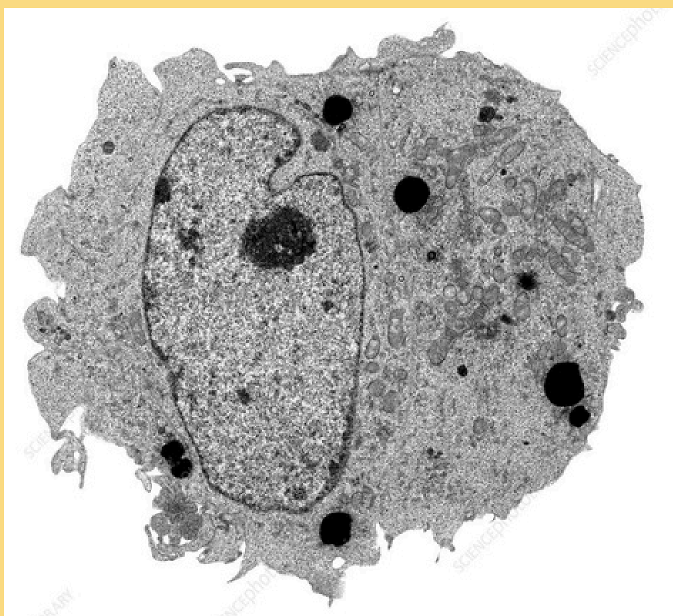
Tasmanian Devil with DFTD

What Is Devil Facial Tumor Disease (DFTD)?

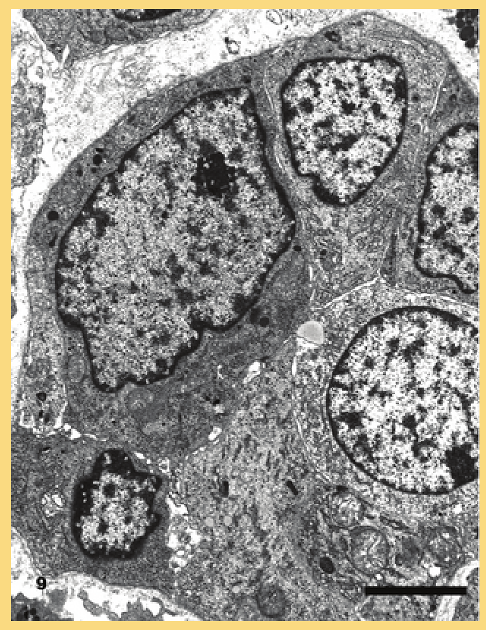
DFTD is an **aggressive**, **non-viral**, **clonally transmissible cancer** which affects **Tasmanian devils**, a marsupial native to **Tasmania**

Differences Between DFTD and Normal Cancer

Criteria	"Normal" Cancer Cells	DFTD
How it arises	Originates from an **organism's own cells,** arising from multiple **DNA mutations**	Originated from a "**founder**" in **1986**, cancer is spread through bites resulting in **allogeneic** transfer of cells
Immune Evasion	**Neoantigens** are present on **MHC Class I** molecules on cancer cells Healthy T cells detects these antigens using cancer **immuno-survelliance**	**Downregulation** of genes for antigen presentation → DFTD cells lack MHC Class I molecules on its surface → T cells **can't detect**
Evolution	Dies with host, **no long-term evolutionary continuity**	Evolve via **epigenetic modifications**, with no change to DNA, but rather **reversible and heritable changes** to gene activity



TEM Scanning of a Lung Cancer Cell



TEM Scanning of a DFTD Cell

Different Cell Strains

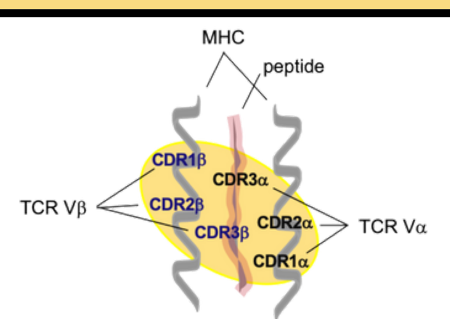
Criteria	DFT1 (**Originated** from a **female**)	DFT2 (**Originated** from a **male**)
Year of Emergence	**1996**	**2014**
Geographic Distribution	**Spread** across **Tanzania;** reached **epidemic proportions**	**Confined** to the **D'Entrecasteaux Peninsula**; not **widespread**
Tumour cell strains are **phenotypically similar**		

Immunogenetics

Low **MHC variation** in Tasmanian Devil Population

- Immune systems **can't distinguish** between **own cells** & **genetically similar tumor** cells from **another** devil
- Tumour **doesn't appear foreign** enough to trigger a **rejection** response → **fails** to **recognize allografts**

(allografts: tissue from another individual)



MHC (Major Histocompatibility Complex), a group of protein molecules to recognize and present antigens

MHC Variations

Cytogenetics

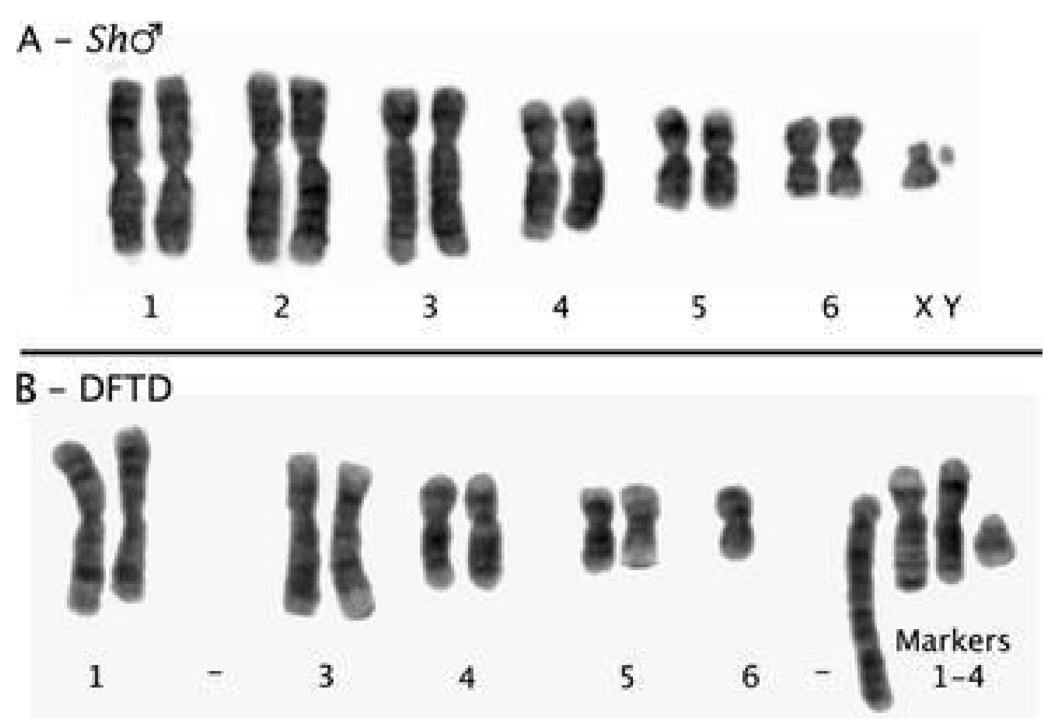
Aneuploidy (abnormal number of chromosomes) → cells susceptible to the accelerated acquisition of mutations → cancer

Loss of:

- both copies of chromosome 2
- one copy of chromosome 6
- both sex chromosomes

Addition of:

- four unidentified marker chromosomes



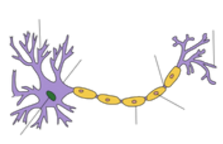
Pathogenesis

DFTD tumors are locally invasive: regional to the **lymph node**
Highly **metastatic** to **thoracic** and **abdominal** viscera (65% of cases)
Rapid disease progression, 100% mortality rate

- large primary tumours obstruct feeding → starvation
- complications associated with metastasis

Shares gross pathological and immunophenotypic features with human **Merkel cell carcinoma**
Expresses genes characteristic of **Schwann cells** or its precursors

Merkel cell carcinoma



Schwann cells

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